# Parent-Level Debt and Subsidiary-Level Risk-Taking: Evidence from Large Bank Holding Companies *

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**Abstract**

We investigate the influence of the parent company's capital structure of a bank holding company's (BHC) on the risk-taking behavior of its subsidiary banks, taking into account the BHC's organizational structure. Our analysis shows that increased parent debt is associated with reduced risk-taking by subsidiary banks. This effect is particularly prominent in BHCs with significant non-bank business, which would give the parent company greater incentives to constrain risk-taking in the subsidiary bank to prevent any spillover of loss. These findings highlight the need to consider the capital structures of the subsidiary, the parent, and the overall BHC—as well as the organizational structure—when assessing a subsidiary bank's risk profile.

**Keywords:** bank holding company, parent-level debt, bank subsidiary, risk management

**JEL Codes:** G21, G28, G32, G33, G38

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# 1 Introduction

Since Jensen and Meckling [1976], corporate finance has a long tradition of considering the incentive implications of capital structure; the study of commercial banks makes no exception. Numerous empirical studies examine the impact of banks' capital structure on their risk-taking incentives, and virtually all studies explaining bank risk-taking would include bank capitalization as a control variable. Yet most modern commercial banks operate within a bank holding company (BHC), raising an essential question: which entity's capital position matters? A BHC can have several relevant balance sheets: the commercial bank's own, those of other subsidiaries, the parent company's, and the consolidated balance sheet of the whole group's. While prior research typically focuses on the capitalization of the commercial bank *or* that of the consolidated BHC, this paper argues that one needs to take a wholistic view, emphasizing the capital position of the parent company. Specifically, we study how the *parent debt ratio*, defined as the share of a BHC's non-deposit debt issued by its parent company, affects the risk-taking of its commercial bank subsidiaries.

Avraham et al. [2012] provide an insightful overview of the organizational structure of US bank holding companies. Figure 1 reproduces their findings for a typical large BHC, in which the holding company simultaneously controls a bank subsidiary, a non-bank subsidiary, and a commercial bank.[1] This multi-tiered organization corresponds to a multi-tiered capital structure. Furthermore, the multi-level capital structure, as proxied by the allocation of external debt, also shows significant time-series and cross-sectional variation: between 2012 and 2022, the average parent debt ratio rose from 4.9% to 12.6% (Figure 2). In 2021, 33 of the 123 BHCs in our sample had no parent-issued debt, while

[1]Some BHCs also own foreign bank and non-bank subsidiaries.

in the top half of the distribution the average parent debt ratio was 29.65%. Despite such variation, the effects of parent-level debt on subsidiary bank behavior remain unexplored. We address this gap by examining how the parent debt ratio influences subsidiary banks' risk-taking—an important question given that commercial banks account for the bulk of BHC assets, rising from 89.03% in 2012Q1 to 94.45% in 2022Q4 (Figure 3).

We find that a higher parent debt ratio is associated with decreased risk-taking by subsidiary banks. Specifically, an increase in the parent debt ratio correlates with lower risk-weighted leverage (measured as risk-weighted assets to tier 1 capital), reduced leverage, and a higher subsidiary bank z-score. These findings suggest that higher levels of parent debt can lead to reduced risk-taking within subsidiary banks.

The underlying mechanism is as follows: if the parent company issues less debt, the subsidiary bank may need to increase its own debt issuance to finance investments, potentially incentivizing greater risk-taking due to limited liability. Under limited liability, the downside is shared with debt holders, while shareholders pocket the upside. However, if the parent company issues more debt and injects the proceeds as equity into its subsidiaries, the parent debt acts as a contractual commitment that mitigate the subsidiary bank's risk-taking behavior.

Similar reasoning extends to the case where the parent issues little debt but there is cross-guarantee. The cross-guarantee provision of Financial Institutions Reform, Recovery, and Enforcement Act of 1989 (FIRREA) permits the Federal Deposit Insurance Corporation (FDIC) to charge off any expected losses related to the failure of one subsidiary bank of a multi-bank holding company to the capital of a related subsidiary bank within the same multi-bank holding company (Ashcraft [2005]). This implies that BHCs with multiple bank subsidiaries may engage in less risk-taking, as they must use their own capital to rescue failing subsidiaries before FDIC intervention. Our findings confirm that subsidiary banks affiliated with multi-bank BHCs take on less risk when the parent company issues more debt.

Our paper highlights the function of internal capital markets within bank holding companies (BHCs). Unlike existing studies—some of which focus on risk-sharing through internal capital markets among global banks (e.g., Cetorelli and Goldberg [2012])—we emphasize the role of internal capital markets in shaping the risk-taking incentives of domestic subsidiary banks within U.S. BHCs.

While the empirical evidence is suggestive, a major concern in the identification of the effect of parent debt on the subsidiary bank's risk-taking is reverse causality: parent company issues less debt due to the risk-taking behavior of subsidiary bank. We address this identification concern by using, as a natural experiment, the COVID-19 outbreak in 2020Q1, which led to an exogenous increase in parent debt issuance. Under this identification, after 2020Q1, the reason for parent debt issuance is not endogenous but instead exogenous. Following the COVID-19 outbreak, parent debt issuance was driven by external reasons unrelated to subsidiary bank risk-taking, mitigating concerns about endogeneity. This allows us to establish a causal relationship between parent debt and subsidiary bank risk-taking. We implement a difference-in-differences (DID) analysis, defining treatment and control groups. The control group consists of BHCs with a non-bank asset ratio of zero, while the treatment group includes BHCs with a non-bank asset ratio above the 90th percentile threshold before the pandemic. Our DID results confirm that, following the COVID-19 outbreak, BHCs with high non-bank asset ratios significantly reduced subsidiary banks' risk-taking. To further sharpen identification, we conduct a subsample analysis around the COVID-19 outbreak, leveraging the sharp increase in parent debt issuance during this period. Our results confirm that a higher parent debt ratio reduces subsidiary bank risk-taking, reinforcing the robustness of our findings and mitigating concerns about sample selection bias.

We present a simple model that captures our empirical finding. In the model, the BHC comprises both a bank subsidiary and a non-bank subsidiary. The parent company issues debt and allocates the proceeds as equity investments into these subsidiaries. The bank subsidiary faces investment opportunities that involve a trade-off between risk and return. our model predicts that when the parent company issues more debt and invests the proceeds as equity into its subsidiaries, it can mitigate the subsidiary bank's risk-taking behavior. Conversely, if the parent company issues less debt, the subsidiary bank may need to increase its own debt issuance, which can incentivize greater risk-taking. This occurs because, under limited liability, the downside is shared with debt holders, while shareholders pocket the upside.

This study contributes to the large literature on the financial ratio regulations aimed at curbing financial institutions' risk-taking incentives. The specific ratio examined here is the parent company's capital structure, which differs from prior research that primarily focus on capital structure at the commercial bank level or the consolidated BHC level (e.g., Keeley [1990], Shrieves and Dahl [1992], Demsetz and Strahan [1997], and Ellul and Yerramilli [2013]).

This paper also contributes to the literature on the organizational structure of financial institutions and risk-taking. Our analysis focuses on the multi-level organizational structures of BHCs and their distinct balance sheets—particularly that of the parent company—rather than the single balance sheet of integrated conglomerates emphasized by Freixas et al. [2007]. Moreover, we highlight the importance of considering existing BHC organizational structures when assessing risk-taking, in contrast to Kahn and Winton [2004] and Luciano and Wihlborg [2018], who design optimal structures to mitigate risk-taking. Closest to our approach is Li et al. [2024], who argue that a bank's organizational structure should be evaluated alongside its capital structure and bankers' compensation contracts when discouraging banker's risk-taking.

Finally, we add to the nascent literature on parent debt in BHCs, where existing research is scarce. Tucker et al. [2023] show that the parent-only leverage ratio is positively associated with the parent's credit default swap premium, suggesting that parent-only balance sheet data can inform assessments of parent credit risk and that consolidated reporting can obscure critical entity-level details. Building on these insights, we examine how parent company debt affects subsidiary bank risk-taking, offering a novel perspective on risk transmission within BHCs.

The paper is structured as follows: Section 2 describes the data sources, key variables, and summary statistics. Section 3 conducts the empirical analysis on the relationship between parent debt and subsidiary banks' risk-taking. Section 4 presents the difference-in-differences analysis. Section 5 examines the impact of the cross-guarantee mechanism. Section 6 introduces the theoretical model. Section 7 conducts robustness check. Finally, Section 8 concludes.

## 2 Data and Summary Statistics

In this section, we describe our data sample, outline the construction of the parent debt ratio and subsidiary banks' risk-taking measures, and summarize our sample. Table 1 provides detailed definition of these variables.

## 2.1 Data Sources

### 2.1.1 BHC-level Data

Our BHC-level data are sourced from the Consolidated Financial Statements for Holding Companies (FR-Y9C) and Parent Company Only Financial Statements for Large Holding Companies (FR-Y9LP) provided by the Federal Reserve Bank. We use data spanning from 2012 to 2022, which include quarterly observations of the income statement and balance sheets of all U.S. BHCs and their parent-only companies. We drop BHCs that do not file the FR-Y9C or FR-Y9LP for each quarter during the entire sample period to avoid the impact of BHCs' joinning or exiting on our empirical results. Finally, we collect data for 123 BHCs over the sample period.

### 2.1.2 Bank-level Data

The subsidiary bank-level data are from the U.S. Call Reports provided by the Federal Reserve Bank, covering the period from 2012 to 2022. This dataset contains quarterly observations of the income statement and balance sheets of all U.S. commercial banks, some of which are the subsidiaries of BHCs. The data contain bank-level identifier (RSSD9348) that can be used to link to the BHC-level data. Finally, our balanced panel consists of 157 unique bank subsidiaries.

## 2.2 Parent Debt Ratio

Our key explanatory variable is the parent debt ratio, defined as the parent's non-deposit liability scaled by the BHC's non-deposit liability. The parent's non-deposit liability includes the parent company's securities sold under agreements to repurchase, borrowings of commercial paper and other borrowings with a remaining maturity of one year or less, other borrowed money with a remaining maturity of more than one year, and subordinated notes and debentures. BHC's non-deposit liability contains group-level consolidated total liabilities but excludes deposits in domestic offices, foreign offices, Edge and Agreement subsidiaries, and international banking facilities (IBFs).

## 2.3 Subsidiary Bank's Risk-taking

To better measure bank's risk, we mainly focus on subsidiary bank's credit risk from both the asset and liability side. We also use Z-Score to measure bank's risk, defined as the return on assets plus equity-to-asset ratio, scaled by the standard deviation of return on assets, where the equity-to-asset ratio is calculated as (book equity)/(total assets), thus, Z-Score comprises risk measure from both the liability and asset sides. A higher Z-Score indicates greater bank stability. Due to the skewness of Z-Score, we use its natural logarithm, which follows a normal distribution (Laeven and Levine [2007]). Following Li and Song [2022], we also use the risk-weighted leverage, defined as a bank's risk-weighted asset divided by tier-1 capital as our risk-taking measure. The risk-weighted leverage is the product of a bank's asset risk weight and leverage and thus summarizes a bank's risk-taking on both margins, therefore, we also include both asset risk weight and leverage as our dependent variables. We winsorize all the variables at the 1% level to minimize the impact of outliers.

## 2.4 Control Variables

We use market data from the Center for Research in Security Prices (CRSP) and regulatory accounting data from Call Reports, FR Y-9C, and FR Y-9LP to construct bank- and BHC-level controls that are considered as important determinants of a bank's risk-taking behavior.

These control variables include bank size ($LnTA$), leverage ratio ($Debt/Equity$), and Tobin's q of the BHCs ($TobinQ$). Keeley [1990] shows that Tobin Q, as a proxy of market power or charter value, is negatively related to a bank's risk-taking. Boyd and Runkle [1993] suggest a negative relationship between bank size and bank risk. Acosta-Smith et al. [2020] find that a leverage ratio requirement associated with the introduction of the Basel III Leverage Ratio can incentivize banks bound by it to increase their risk-taking.

## 2.5 Summary Statistics

Table 2, Panel A, summarizes the sample statistics for the 2012-2022 period. The parent debt ratio averages 11.10%, ranging from 0 to 75.6%. The mean logarithm of asset size is 15.53, which translates to a mean of USD 5.53 billion in total assets. The average Tobin's Q for BHCs is 1.04.

Regarding subsidiary banks' risk-taking measures, the mean logarithm of the Z-score is 3.55, the average risk-weighted leverage is 6.86, the risk weight is 0.67, and the leverage ratio is 9.84.

# 3 Empirical Approach and Results

The central premise of our paper is the impact of parent debt ratio on the risk-taking of subsidiary banks. In this section, we present evidence that establishes the negative link between the parent debt ratio and subsidiary bank's risk-taking.

For each subsidiary bank, we estimate the following model:

$$\begin{aligned} RiskTaking_{ijt} = \alpha + \beta ParentDebtRatio_{j,t-4} + \\ + \gamma Controls_{i,t-4} + \delta TobinQ_{j,t-4} + \lambda_i + \eta_t + \epsilon_{ijt} \end{aligned} \tag{1}$$

where, for each subsidiary bank $i$ affiliated with BHC $j$ and quarter $t$, $ParentDebtRatio$ is calculated as the parent's non-deposit liability divided by the BHC's non-deposit liability, $Controls$ represents a set of bank-level control variables described in Section 4.4. $TobinQ$ is a BHC-level variable. Since most subsidiary banks are not public, we could not calculate the bank-level Tobin Q. Therefore, we use BHC-level Tobin Q as an alternative in our regression model. All explanatory variables are lagged by four quarters to mitigate concerns over look-ahead bias. $RiskTaking$ is a measure of bank's risk-taking originating in quarter $t$. In all models, standard errors are clustered at subsidiary bank level. We control for subsidiary bank fixed effect to make our regression result robust.

Table 3 presents the empirical evidence showing a negative relationship between the parent debt ratio and the subsidiary bank's risk-taking during the 2012-2022 period. Specifically, for a one standard deviation increase in the parent debt ratio (0.129), we observe an expected 4.86% increase in Z-Score (0.3766 x 0.129 x 100% = 4.86%), a

15.90% decrease in risk-weight leverage (-1.2323 x 0.129 x 100% = -15.90% ), and a 20.23% decrease in leverage (-1.5685 x 0.129 x 100% = -20.23% ).

# 4 Identification Strategy:Difference-in-Differences

The empirical results in section 4 show a negative relationship between parent debt and subsidiary bank's risk-taking, but this finding may be tainted by endogenous concerns. Specifically, the risk-taking activities of the subsidiary banks could potentially influence the parent company to raise additional external debt in order to mitigate the adverse effects of subsidiary bank's risk-taking. To establish a causal link between parent debt ratio and the subsidiary bank's risk-taking, we employ an instrumental variable (IV) that creates plausibly exogenous variation in parent debt ratio. Parent debt ratio are more likely to be affected by the breakout of COVID-19. Therefore, we instrument the BHC's parent debt ratio with the timing of COVID-19 outbreak. COVID-19 began impacting the U.S. BHCs from the second quarter of 2020, inducing a time-serious variation in parent debt ratio during our sample period. Moreover, the outbreak of COVID-19 is an exogenous event, unlikely to be affected by the BHC's financing policy. A critical assumption of this IV approach is that COVID-19 affects subsidiary bank's risk-taking only through its impact on parent debt ratio and not through other mechanisms. Since COVID-19 has impacted all BHCs in the U.S., to disentangle the different effects of parent debt on bank subsidiary's risk-taking across BHCs, we implement a difference-in-differences (DID) analysis by defining treatment and control groups. BHCs typically have both bank and non-bank subsidiaries. If the proportion of non-bank assets within the BHC is high, the parent may have greater incentive to reduce subsidiary bank's risk-taking when issuing parent debt. This is because, if subsidiary bank takes excessive risk and fails, the parent may need to utilize the non-bank assets to bail in subsidiary bank, potentially affecting the non-bank's profit since the non-bank's profit is usually higher (Lu et al. [2023]). Based on this rationale, the control group includes BHCs without any non-bank assets during our sample period, while the treatment group contains BHCs, whose non-bank asset ratio is above the 90th percentile threshold of all BHCs prior to the COVID-19 pandemic.

The difference-in-differences specification is as follows:

$$RiskTaking_{ijt} = \alpha + \beta_1 Treated + \beta_2 Post + \beta_3 Treated * Post + \gamma Control_{i,t-4} + \lambda_i + \epsilon_{ijt}, \tag{2}$$

where $i$, $j$ and $t$ denote the subsidiary bank, BHC, and quarter, respectively. The variable $Post$ equals one for the post-COVID period and zero for the pre-COVID period. $Treated$ is equal to one for the treated sample and zero for the control sample. The difference-in-differences estimator, $\beta_3$, is the coefficient on the product of the $Treated$ and $Post$.

Bertrand et al. [2004] argue that the standard errors of a pooled difference-in-differences estimator are generally underestimated. To address this concern, they recommend aggregating each firm's pre- and post-treatment data into a single pre- and single post-observation. Following Bertrand et al. [2004] and Pogach and Unal [2019], we construct risk-taking measures for each subsidiary bank in the pre- and post-COVID periods.

Table 2, Panel B, presents the statistics of the reduced sample from 2017 to 2022. The parent debt ratio averages 17.00%, while the mean logarithm of asset size is 15.98.

The average Tobin's Q for BHCs is 1.04. Compared with the whole sample, both the mean parent debt ratio and asset size increased slightly during this period.

Table 4 presents the regression results for the difference-in-differences analysis. The coefficient of interest on the interaction term shows that the logarithm of Z-Score for the treated group rose 33.84 percentage points in the post-COVID period relative to the control group, significant at the 5 percent threshold. The effect is robust to the inclusion of control variables.

# 5 The Impact of Cross-guarantee on Subsidiary Bank's Risk-taking

In our previous analysis, we focus on that the parent debt reduces its subsidiary bank's risk-taking. The reason may be joint liability. Specifically, if the parent defaults, the subsidiary bank is also liable for the parent's debt, creating a form of contractual bail-in. Next, we explore the regulatory bail-in mechanisms. The cross-guarantee provision of Financial Institutions Reform, Recovery, and Enforcement Act of 1989 (FIRREA) permits the Federal Deposit Insurance Corporation (FDIC) to charge off any expected losses related to the failure of one subsidiary bank of a multi-bank holding company to the capital of a related subsidiary bank within the same multi-bank holding company (Ashcraft [2005]). We illustrates this with an example: suppose there are two BHCs, A and B. BHC A has three bank subsidiaries - A1, A2, A3 while BHC B only has one bank subsidiary - B1. If A1 defaults, the FDIC can require BHC A to use the capital from A2 and A3 to cover A1's debt. However, if B1 defaults, BHC B has no other bank subsidiaries to draw capital from, necessitating an FDIC bailout. From this, we infer that BHCs with multiple bank subsidiaries may engage in less risk-taking, as they must first use their own capital to rescue a defaulted subsidiary before the FDIC steps in. Therefore, we test whether BHCs with multiple bank subsidiaries indeed reduce subsidiary bank's risk-taking.

For each bank subsidiary, we estimate the following model:

$$\begin{aligned} RiskTaking_{ijt} = \alpha + \beta_1 ParentDebtRatio_{j,t-4} + \beta_2 Multibank_{j,t} + \\ + \beta_3 Multibank_{j,t} * ParentDebtRatio_{j,t-4} + \\ + \gamma Control_{i,t-4} + \delta TobinQ_{j,t-4} + \lambda_i + \eta_t + \epsilon_{ijt} \end{aligned} \quad (3)$$

where, for each bank subsidiary $i$ affiliated with BHC $j$ and in quarter $t$, $Multibank$ equals to one if the BHC has more than one subsidiary banks, otherwise, zero. $Control$ and $TobinQ$ are the same as these we used in regression model (1). Table 2, Panel D, summarizes key variables used in the model.

Table 5 demonstrates a negative relationship between the cross-guarantee and subsidiary bank's risk-taking. For a one standard deviation increase in parent debt ratio (0.129), we observe an expected additional 21.66% decrease in the risk-weighted leverage (0.129 x -1.6793 x 100% = -21.66% ) and a 2.53% decrease in the risk weight (0.129 x -0.1958 x 100% = -2.52% ) if the BHC has more than one subsidiary banks.

# 6 Robustness Check

The robustness check addresses concern regarding whether our results are truly driven by sample selection. Table 2, Panel D, summarizes the sample statistics for the 2017-2022 period. On average, the parent debt ratio is 13.7%. The mean logarithm of asset size is 15.76. The average debt/equity ratio is 7.24. The average Tobin's Q for BHCs is 1.04.

In the difference-in-differences analysis, we use sample from 2017 to 2022. For the robustness check, we limit our sample to this same period, that is, 2017-2022, Table 6 shows a negative relationship between the parent debt ratio and the subsidiary bank's risk-taking. Specifically, for a one standard deviation increase in the parent debt ratio (0.139), we observe an expected a 5.08% increase in Z-Score (0.3655 x 0.139 x 100% = 5.08%), a 15.90% decrease in risk-weighted leverage (-0.5062 x 0.139 x 100% = -15.90% ), and a 20.23% decrease in leverage (-1.7158 x 0.139 x 100% = -20.23% ).

# 7 A Model of Parent Debt and Subsidiary Bank's Risk-taking

We present a model to explain the relationship between parent debt and the subsidiary bank's risk-taking. Detailed derivations and proofs are provided in the Appendix. The model operates over two time periods, denoted as t = 0 and t = 1. We consider a BHC wholly owning a banking subsidiary and a non-bank subsidiary within its organizational structure.

## 7.1 t = 0

At t = 0, the consolidated balance sheet of BHC comprises one unit of asset, $D_c$ units of debt, and $E_c$ units of equity. The BHC also submit a parent company only balance sheet to the Federal Reserve. The parent company only has $E_p$ units equity and $D_p$ units debt, and capitalizes the banking and non-bank subsidiaries pro rata, i.e., injecting $\alpha(E_p+D_p)$ into the non-bank subsidiary and $(1-\alpha)(E_p+D_p)$ into the banking subsidiary in the form of equity, with $\alpha$ and $(1-\alpha)$ being the relative sizes of the non-bank and banking subsidiaries respectively. Here, under this assumption,

$$E_p + D_p \tag{1}$$

is the *equity capital ratio of both subsidiaries from the parent.* This can be a *strong* assumption since the capital allocation should be endogenous. A hand-waving justification can be that the capital ratios are set binding for both the banking and non-bank subsidiary. The non-bank subsidiary's assets, debt, and equity are denoted as $\alpha$, $D_n$, and $E_n$, respectively. Similarly, the bank subsidiary's assets, debt, and equity are represented as 1-$\alpha$, $D_b$, and $E_b$, respectively. We assume that $E_b$ and $E_n$ are fixed. The balance sheets for the BHC, parent company only, nonbank subsidiary, and bank subsidiary are presented in Table 1. From these balance sheets, we observe that $E_b$ equals $A_b$ and $E_n$ equals $A_n$.

Table 1: Balance Sheets at t = 0

BHC

| | |
|---|---|
| 1 | $D_c$ |
| | $E_c$ |

Parent

| | |
|---|---|
| $A_b$ | $E_p$ |
| $A_n$ | $D_P$ |

Nonbank Subsidiary

| | |
|---|---|
| $\alpha$ | $E_n$ |
| | $D_n$ |

Bank Subsidiary

| | |
|---|---|
| 1-$\alpha$ | $E_b$ |
| | $D_b$ |

At t = 0, the subsidiary bank encounters an investment opportunity that yields a payoff $R_b$ with probability of P($R_b$) and a payoff of S with probability of 1-P($R_b$). $S$

denotes the salvage value. $\mathrm{P}(R_b)$ is a decreasing function in $R_b$, indicating that higher payoffs are less likely. The two-point distribution of the investment is

$$R = \begin{cases} R_b, & \text{with probability } P(R_b), \\ S, & \text{with probability } 1 - P(R_b), \end{cases}$$

We also assume that the loan loss is sufficiently large:

$$S < 1 - E_p. \tag{2}$$

The assumption states that the possible loss from loan portfolio *will* result in the insolvency of the subsidiary bank if there is no capitalization financed by parent's debt. That is, $S(1-\alpha) < (1-\alpha)(1-E_p-D_p)$ when $D_p = 0$. The alternative assumption $S \geq 1 - E_p$ suggests that the subsidiary bank never goes bankruptcy because the equity capital within BHC is in abundance and sufficient to cover the loss of the loan, which would lead to an uninteresting situation where limited liability is never triggered. To capture the risk-return trade-off, we assume the probability of success is

$$P(R_b) = 1 - \beta R_b. \tag{3}$$

The assumption parameterizes $P(R_b)$ and takes a linear form, which can probably be generalized. The return for the nonbank asset is $R_n$. We assume that the investment of the nonbank subsidiary is risk-free, denoted by $r_f$. This is a strong assumption, but we may obtain similar results under the assumption of $i.i.d.$ returns and diversification because we consider only the ex ante risk-taking decision. A hand-waving justification can be that certain nonbank activities such as advising, market-making, asset management, and security underwriting are supposed to be risk-free. We normalize the risk-free rate to 1. Since subsidiary bank do not impose negative rate on deposits, we have

$$r_b = r_f = 1. \tag{4}$$

The assumption provides a simple starting point, where subsidiary bank would not gain on its funding side due to their possible deposit market power.

## 7.2 t = 1

At t = 1, the balance sheets of non-bank and bank subsidiaries are as follows:

Table 2: Balance Sheets at t = 1

| Nonbank Subsidiary | | Bank Subsidiary (Succeed) | | Bank Subsidiary (Fail) | |
|---|---|---|---|---|---|
| $R_n\alpha$ | $R_n\alpha - D_n$ | $R_b(1-\alpha)$ | $R_b(1-\alpha) - D_b$ | $S(1-\alpha)$ | $\max\left\{S(1-\alpha) - D_b, 0\right\}$ |
| | $D_n$ | | $D_b$ | | $D_b$ |

The shareholder of the BHC solves the following program:

$$\begin{aligned} \max_{R_b} L = {} & P(R_b)[R_n\alpha - D_n + R_b(1-\alpha) - D_b - D_p] + \\ & (1 - P(R_b))[R_n\alpha - D_n + \max\left\{S(1-\alpha) - D_b, 0\right\} - D_p] \end{aligned} \tag{5}$$

where $D_b = (1-\alpha)(1 - E_p - D_p)$, and $D_n = \alpha(1 - E_p - D_p)$. The first-best risk level is determined by maximizing the expected payoff for a given production technology, independent of the capital structure. After taking the first-order derivative, it yields the first-best solution:

$$R_b^{FB} = \frac{1 + \beta S}{2\beta}. \tag{6}$$

The subsidiary bank fails if and only if $S(1-\alpha) - (1-\alpha)(1-E_p-D_p) < 0$, or equivalently,

$$D_p < (1-E_p) - S \equiv \widehat{D}'_p \tag{7}$$

When the subsidiary bank fails, the parent/BHC survives if and only if the equity value of the non-bank subsidiary allows for the repayment of parent debt. That is, $[R_n\alpha - \alpha(1-E_p-D_p)] - D_p \geq 0$, or equivalently,

$$D_p \leq \frac{\alpha[R_n - (1-E_p)]}{1-\alpha} \equiv \widehat{D}''_p. \tag{8}$$

We have two broad cases (1) no parent/BHC bankruptcy when the bank fails, and (2) parent/BHC fails. When the parent/BHC fails, the subsidiary bank can fail or not.

## (1) No Parent Bankruptcy

We start with a case where $D_p \leq \widehat{D}''_p$, so that the BHC does not fail when its banking subsidiary incurs the loan losses.

**Lemma 1.** *If and only if $\alpha > \frac{(1-E_p)-S}{R_n-S} \equiv \widehat{\alpha}$, it holds that $\widehat{D}'_p < \widehat{D}''_p$. Excessive risk-taking occurs when $D_p < \widehat{D}'_p$, whereas the first-best risk level emerges when $D_p \in \left[\widehat{D}'_p, \widehat{D}''_p\right]$.*

See Appendix A1: Proof of Lemma 1

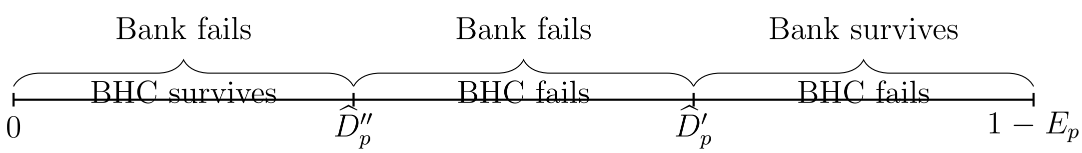
When $D_p < \widehat{D}'_p$, the subsidairy bank fails in distress whereas the BHC survives. The limited liability at the subsidiary bank creates incentives for risk-taking when the risk-taking decision is made by the shareholder of the BHC.

When $D_p \in \left[\widehat{D}'_p, \widehat{D}''_p\right]$, both the subsidiary bank and the parent/BHC survive, and the loan losses are fully passed onto the shareholder of the BHC. The absence of limited liability distortion on the bank level align the private risk-taking decision with the first-best level.

**Lemma 2.** *A more significant non-banking subsidiary, i.e., $\alpha$ exceeds $\widehat{\alpha}$, reduces risk-taking for a given $D_p$.*

A higher $\alpha$ increases $\widehat{D}_p''$, while $\widehat{D}_p'$ stays unchanged, creating/expanding the range where the first-best risk level emerges. Intuitively, when $\alpha$ approaches 0 and the equity value from the non-bank practically disappears, the limited liability distortion is fully relayed onto the parent/BHC level, creating excessive risk-taking incentives.

**Lemma 3.** *If and only if $\alpha < \frac{(1-E_p)-S}{R_n-S} \equiv \widehat{\alpha}$, it holds that $\widehat{D}_p' > \widehat{D}_p''$. Excessive risk-taking occurs when $D_p < \widehat{D}_p''$ if the parent/BHC does not fail.*




## (2) Parent Bankruptcy Risk

### (2.1) Both the parent/BHC and bank subsidiary fail

**Lemma 4.** *When both the parent/BHC and subsidiary bank fail, excessive risk-taking still occurs because limited liability distortion emerges at the subsidiary bank level.*

$R_b = \frac{1+\beta(1-E_p-D_p)}{2\beta}$

See Appendix A1: Proof of Lemma 4

Subsidiary bank still takes excessive risk because limited liability still kicks in at the bank level.

### (2.2) The parent/BHC fails but subsidiary bank not

**Lemma 5.** *When the parent/BHC fails but the subsidiary bank does not, it yields the first best solution with*

$$R_b^{FB} = \frac{1 + \beta S}{2\beta}. \tag{9}$$

See Appendix A1: Proof of Lemma 5

When the parent/BHC fails but the subsidiary bank does not, the absence of limited liability distortion on the bank level align the private risk-taking decision with the first-best level.

## 8 Conclusion

We find that parent debt reduces subsidiary banks' risk-taking. Using the COVID-19 outbreak as a natural experiment, we confirm the robustness of our results. We attribute this effect to the cross-guarantee mechanism, which explains why higher parent debt leads to lower risk-taking by subsidiary banks. When a BHC has multiple bank subsidiaries, the parent's increased debt issuance helps mitigate the risk of its subsidiary banks.

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**Table 1**

**Variable definitions and sources of data used in the study.** This table summarizes the variable definitions and sources of data used in our study.

| Variables | Definition |
|---|---|
| Tobin Q | Tobin Q is the group-level consolidated tobin's q, and defined as the ratio between the market value of BHC (MV) over the replacement cost of its assets (RC). It is estimated by Tobin Q = (AT + ME - BE) / AT, where AT is total asset, ME is the market value of equity at year end by multiplying stock price with the common shares outstanding, BE is (Stockholders Equity + Deferred Taxes + Investment Tax Credit - Preferred Stock). I use the code from WRDS to calculate it. Please see the link for details: https://wrds-www.wharton.upenn.edu/pages/wrds-research/applications/risk-and-valuation-measures/tobins-q-altman-z-score-and-companys-age/ |
| Z-Score | Z-score is a proxy for bank stability. It is estimated as return on assets plus equity-to-asset ratio, scaled by the standard deviation of return on assets, where equity-to-asset ratio is estimated by (book equity)/(total assets). The variable is calculated at the annual level. To avoid the skewness, we take the logarithm of it. |
| ROA | ROA is estimated as net income divided by total assets: RIAD4340(bank subsidiary's net income (loss) attributable to bank from FFIEC041)/ RCON2170(bank subsidiary's total asset from FFIEC 041) or RIAD4340(bank subsidiary's net income (loss) attributable to bank from FFIEC031)/RCFD2170 (bank subsidiary's total asset from FFIEC 031) |
| SD ROA | Standard deviation of subsidiary bank 's return on asset is a proxy for bank riskiness behavior. It is estimated as the standard deviation of the quarterly return on assets in a given year. The variable is calculated at the annual level. |
| Debt/Equity | Debt/Equity is the ratio of commercial bank's total liability divided by its total equity: RCON2948(bank subsidiary's total liability from FFIEC 041)/(RCON2170(bank subsidiary's total asset from FFIEC 041) - RCON2948(bank subsidiary's total liability from FFIEC 041) ) or RCFD2948(bank subsidiary's total liability from FFIEC 031)/(RCFD2170 (bank subsidiary's total asset from FFIEC 031) -RCFD2948 (bank subsidiary's total liability from FFIEC 031) ) |
| Ln AT | Ln AT is the logarithm of subsidiary bank 's total assets: ln(RCON2170(bank subsidiary's total asset from FFIEC 041)) or ln(RCFD2170((bank subsidiary's total asset from FFIEC 031)) |

| | |
|---|---|
| Parent Debt Ratio | Parent Debt Ratio is the ratio of the parent company's non-deposit debt liabilities to BHC's non-deposit liability: quarterly (BHCP2309(parent company's borrowings of commercial paper with a remaining maturity of one year or less) + BHCP2332(parent company's other borrowings with a remaining maturity of one year or less) + BHCP0368(parent company's other borrowed money with a remaining maturity of more than one year) + BHCP4062(parent company's subordinated notes and debentures)+ BHCP0279(parent company's securities sold under agreements to repurchase) )/(BHCK2948(group-level consolidated total liabilities)-BHDM6631(group-level consolidated noninterest-bearing deposits in domestic offices)-BHDM6636(group-level consolidated interest-bearing deposits in domestic offices )-BHFN6631(group-level consolidated noninterest-bearing deposits in foreign offices, Edge and Agreement subsidiaries, and IBFs)-BHFN6636(group-level consolidated interest-bearing deposits in foreign offices, Edge and Agreement subsidiaries, and IBFs )) |
| Multi-bank | If the BHC has more than one subsidiary banks, this value equals to one, otherwise, zero. |
| RW Assets/T1 Capital | Risk-weighted leverage is the ratio of risk-weighted assets to tier 1 capital. It is measured by the RCFAA223(Total risk-weighted assets)/RCFA8274(Tier 1 capital) for banks that file FFIEC031 or the RCOAA223(Total risk-weighted assets)/RCOA8274(Tier 1 capital) for banks that file FFIEC041. |
| RW Assets/Assets | Asset risk weight is measured by the RCFAA223 (Total risk-weighted assets)/RCFD2170(Total asset) for banks that file FFIEC031 or the RCOAA223(Total risk-weighted assets)/RCON2170(Total assets) for banks that file FFIEC041. |
| Assets /T1 Capital | Leverage is measured by RCFD2170(Total assets)/RCFA8274(Tier 1 capital) for banks that file FFIEC031 or RCON2170(Total assets)/RCOA8274(Tier 1 capital) for banks that file FFIEC041. |
| Non-bank Asset Ratio | Non-bank asset ratio is measured by BHCP4778(Total combined nonbank assets of nonbank subsidiaries))/the sum of RCFD2170(Total assets) of all bank subsidiaries within the BHC for banks that file FFIEC031 or BHCP4778(Total combined nonbank assets of nonbank subsidiaries))/the sum of RCON2170(Total assets) of all bank subsidiaries within the BHC that file FFIEC041. |

**Table 2**

**Summary statistics of key characteristics.** This table summarizes the sample statistics of key subsidiary bank-level and BHC-level characteristics used in our study. Panel A uses the full sample of subsidiary banks and BHCs during the 2012-2022 period. Panel B uses the reduced sample for difference-in-differences models over the 2017-2022. Both the full and reduced samples comprise 123 unique BHCs and 157 subsidiary banks.

**Panel A: Full Sample (2012-2022)**

| VARIABLES | (1) N | (2) mean | (3) sd | (4) p10 | (5) p25 | (6) p50 | (7) p75 | (8) p90 | (9) min | (10) max |
|---|---|---|---|---|---|---|---|---|---|---|
| Parent Debt Ratio | 6,904 | 0.111 | 0.129 | 0 | 0 | 0.0608 | 0.211 | 0.299 | 0 | 0.756 |
| ROA | 6,908 | 0.0157 | 0.0477 | 0.00235 | 0.00378 | 0.00706 | 0.0108 | 0.0167 | -0.0878 | 0.888 |
| Debt/Equity | 6,908 | 7.243 | 3.259 | 0.298 | 6.154 | 7.588 | 9.163 | 10.36 | 0 | 32.71 |
| Ln AT | 6,908 | 15.53 | 2.354 | 12.89 | 14.51 | 15.60 | 16.87 | 18.50 | 8.009 | 21.97 |
| Tobin Q | 6,889 | 1.035 | 0.0629 | 0.979 | 1.000 | 1.023 | 1.055 | 1.094 | 0.913 | 1.606 |
| Z-Score | 6,908 | 3.546 | 0.555 | 3.033 | 3.291 | 3.513 | 3.736 | 4.073 | 1.978 | 5.824 |
| RW Assets/T1 Capital | 6,882 | 6.862 | 2.745 | 0.850 | 6.304 | 7.739 | 8.590 | 9.348 | 0.201 | 10.66 |
| RW Assets/Assets | 6,882 | 0.667 | 0.186 | 0.352 | 0.600 | 0.711 | 0.791 | 0.856 | 0.193 | 0.982 |
| Assets /T1 Capital | 6,908 | 9.840 | 3.542 | 2.155 | 9.137 | 10.63 | 11.89 | 13.09 | 1.002 | 16.46 |

**Panel B: Sample for difference-in-differences analysis (2017-2022)**

| VARIABLES | (1) N | (2) mean | (3) sd | (4) p10 | (5) p25 | (6) p50 | (7) p75 | (8) p90 | (9) min | (10) max |
|---|---|---|---|---|---|---|---|---|---|---|
| Parent Debt Ratio | 1,272 | 0.170 | 0.161 | 0 | 0 | 0.137 | 0.296 | 0.366 | 0 | 0.756 |
| ROA | 1,272 | 0.0126 | 0.0273 | 0.00255 | 0.00405 | 0.00732 | 0.0111 | 0.0169 | -0.0530 | 0.399 |
| Debt/Equity | 1,272 | 7.347 | 3.201 | 0.647 | 6.228 | 7.697 | 9.256 | 10.45 | 0.00206 | 17.53 |
| Ln AT | 1,272 | 15.98 | 2.100 | 14.07 | 15.03 | 15.77 | 16.79 | 18.66 | 10.84 | 21.97 |
| Tobin Q | 1,272 | 1.037 | 0.0701 | 0.979 | 0.998 | 1.024 | 1.053 | 1.100 | 0.914 | 1.606 |
| Z-Score | 1,272 | 3.526 | 0.593 | 2.969 | 3.264 | 3.483 | 3.709 | 4.084 | 1.978 | 5.824 |
| SD-ROA | 1,272 | 0.00685 | 0.0130 | 0.00241 | 0.00318 | 0.00393 | 0.00491 | 0.00914 | 0.00106 | 0.126 |
| RW Assets/T1 Capital | 1,260 | 6.713 | 2.662 | 0.772 | 6.420 | 7.632 | 8.279 | 8.977 | 0.201 | 10.66 |
| RW Assets/Assets | 1,260 | 0.649 | 0.199 | 0.296 | 0.576 | 0.696 | 0.791 | 0.852 | 0.193 | 0.982 |
| Assets /T1 Capital | 1,272 | 9.842 | 3.305 | 3.520 | 9.275 | 10.47 | 11.65 | 12.77 | 1.002 | 16.46 |
| Nonbank | 1,272 | 0.321 | 0.467 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
| Time | 1,272 | 0.458 | 0.498 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |

**Panel C: Sample for Cross-guarantee Analysis (2012-2022)**

| VARIABLES | (1) N | (2) mean | (3) sd | (4) p10 | (5) p25 | (6) p50 | (7) p75 | (8) p90 | (9) min | (10) max |
|---|---|---|---|---|---|---|---|---|---|---|
| Parent Debt Ratio | 6,904 | 0.111 | 0.129 | 0 | 0 | 0.0608 | 0.211 | 0.299 | 0 | 0.756 |
| ROA | 6,908 | 0.0157 | 0.0477 | 0.00235 | 0.00378 | 0.00706 | 0.0108 | 0.0167 | -0.0878 | 0.888 |
| Debt/Equity | 6,908 | 7.243 | 3.259 | 0.298 | 6.154 | 7.588 | 9.163 | 10.36 | 0 | 32.71 |
| Ln AT | 6,908 | 15.53 | 2.354 | 12.89 | 14.51 | 15.60 | 16.87 | 18.50 | 8.009 | 21.97 |
| Tobin Q | 6,889 | 1.035 | 0.0629 | 0.979 | 1.000 | 1.023 | 1.055 | 1.094 | 0.913 | 1.606 |
| Z-Score | 6,908 | 3.546 | 0.555 | 3.033 | 3.291 | 3.513 | 3.736 | 4.073 | 1.978 | 5.824 |
| RW Assets/T1 Capital | 6,882 | 6.862 | 2.745 | 0.850 | 6.304 | 7.739 | 8.590 | 9.348 | 0.201 | 10.66 |
| RW Assets/Assets | 6,882 | 0.667 | 0.186 | 0.352 | 0.600 | 0.711 | 0.791 | 0.856 | 0.193 | 0.982 |
| Assets /T1 Capital | 6,908 | 9.840 | 3.542 | 2.155 | 9.137 | 10.63 | 11.89 | 13.09 | 1.002 | 16.46 |
| Multi-bank | 6,908 | 0.363 | 0.481 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |

**Panel D: Reduced Sample (2017-2022)**

| VARIABLES | (1)<br>N | (2)<br>mean | (3)<br>sd | (4)<br>p10 | (5)<br>p25 | (6)<br>p50 | (7)<br>p75 | (8)<br>p90 | (9)<br>min | (10)<br>max |
|---|---|---|---|---|---|---|---|---|---|---|
| Parent Debt Ratio | 3,764 | 0.137 | 0.139 | 0 | 0 | 0.103 | 0.239 | 0.330 | 0 | 0.756 |
| ROA | 3,768 | 0.0172 | 0.0485 | 0.00258 | 0.00406 | 0.00761 | 0.0115 | 0.0177 | -0.0671 | 0.586 |
| Debt/Equity | 3,768 | 7.236 | 3.317 | 0.243 | 6.030 | 7.604 | 9.172 | 10.30 | 0 | 32.71 |
| Ln AT | 3,768 | 15.76 | 2.352 | 13.07 | 14.79 | 15.84 | 17.09 | 18.68 | 8.015 | 21.97 |
| Tobin Q | 3,767 | 1.039 | 0.0669 | 0.979 | 1.001 | 1.026 | 1.059 | 1.101 | 0.914 | 1.606 |
| Z-Score | 3,768 | 3.485 | 0.540 | 3.004 | 3.247 | 3.454 | 3.672 | 3.996 | 1.978 | 5.824 |
| RW Assets/T1 Capital | 3,742 | 6.817 | 2.730 | 0.748 | 6.358 | 7.755 | 8.501 | 9.144 | 0.201 | 10.66 |
| RW Assets/Assets | 3,742 | 0.668 | 0.187 | 0.349 | 0.606 | 0.721 | 0.791 | 0.850 | 0.193 | 0.982 |
| Assets /T1 Capital | 3,768 | 9.715 | 3.447 | 1.792 | 9.236 | 10.55 | 11.68 | 12.69 | 1.002 | 16.46 |

**Table 3**

**Parent debt ratio and subsidiary bank's risk-taking.** This table presents the results from the model for our preliminary analysis used in Section (4). The dependent variables are the logarithm of Z-score (Z-Score), Risk-weighted Leverage (RW Assets/T1 Capital), Asset Risk Weight (RW Assets/Assets) and Leverage (Assets /T1 Capital). The key explanatory variable is parent debt ratio. For each risk-taking measure, Models (1), (3), (5), and (7) contain only parent debt ratio as a control, while Models (2), (4), (6), and (8) contain all set of controls. All models use quarterly data for 2012-2022. All models include both subsidiary bank and quarter fixed effects. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| VARIABLES | Z-Score | Z-Score | RW Assets/T1 Capital | RW Assets/T1 Capital | RW Assets/Assets | RW Assets/Assets | Assets /T1 Capital | Assets /T1 Capital |
| Parent Debt Ratio | 0.5262*** | 0.3766** | -1.5764*** | -1.2323*** | 0.0283 | -0.0019 | -2.6941*** | -1.5685*** |
| | (0.1497) | (0.1466) | (0.3786) | (0.3879) | (0.0337) | (0.0344) | (0.5508) | (0.4479) |
| Tobin Q | | -0.3661 | | 1.7142 | | 0.2252** | | 0.7879 |
| | | (0.4283) | | (1.1271) | | (0.1048) | | (1.2694) |
| Ln AT | | -0.2646** | | 0.2050 | | 0.0059 | | 0.4227*** |
| | | (0.1192) | | (0.1436) | | (0.0182) | | (0.1538) |
| Debt/Equity | | -0.0586*** | | 0.0935** | | -0.0105*** | | 0.3451*** |
| | | (0.0185) | | (0.0364) | | (0.0038) | | (0.0597) |
| Constant | 3.4847*** | 8.3909*** | 7.0423*** | 1.3921 | 0.6658*** | 0.4193 | 10.0980*** | 0.1657 |
| | (0.0158) | (2.1581) | (0.0400) | (2.7717) | (0.0036) | (0.3380) | (0.0580) | (3.0156) |
| Observations | 6,258 | 6,258 | 6,232 | 6,232 | 6,232 | 6,232 | 6,258 | 6,258 |
| R-squared | 0.6881 | 0.7207 | 0.9484 | 0.9505 | 0.9123 | 0.9164 | 0.9226 | 0.9351 |
| Bank_Subsidiary FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |

**Table 4**

**Difference-in-Differences Analysis in Subsidiary Bank's Risk-taking**. This table reports results from difference-in-differences regressions using Equation (2). The treated group includes subsidiary banks with the non-bank asset ratio greater than the 90th percentile before COVID-19, while the control group consists of subsidiary banks that do not have non-bank subsidiary during the whole sample period. Post is an indicator variable equal to zero before COVID-19 and one after. All columns collapse each pre- and post-COVID periods into a single observation for each subsidiary bank. Models (1), (3), (5), and (7) contain Nonbank and Time as controls, while Models (2), (4), (6), and (8) contain all set of controls. All models include subsidiary bank fixed effect. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| VARIABLES | Z-Score | Z-Score | RW Assets/T1 Capital | RW Assets/T1 Capital | RW Assets/Assets | RW Assets/Assets | Assets /T1 Capital | Assets /T1 Capital |
| Treated X Post | 0.3884** | 0.3374** | -0.0955 | -0.3499 | -0.0134 | -0.0498*** | 0.2676 | 0.4710*** |
| | (0.1544) | (0.1301) | (0.1978) | (0.2149) | (0.0208) | (0.0164) | (0.3125) | (0.1388) |
| Post | -0.0893*** | 0.0600 | -0.1090 | 0.1277 | -0.0566*** | 0.0324 | 0.6372*** | -0.6781*** |
| | (0.0309) | (0.1310) | (0.1077) | (0.2170) | (0.0092) | (0.0200) | (0.1607) | (0.1365) |
| Tobin Q | | -3.2102** | | 6.2684 | | 0.8514*** | | -7.3914*** |
| | | (1.5870) | | (3.9176) | | (0.2997) | | (2.5486) |
| Ln AT | | -0.5876 | | -0.6331 | | -0.1185*** | | 0.9709*** |
| | | (0.3545) | | (0.3795) | | (0.0343) | | (0.3366) |
| Debt/Equity | | -0.0649 | | 0.2012* | | -0.0250*** | | 0.9033*** |
| | | (0.0635) | | (0.1156) | | (0.0088) | | (0.0833) |
| Constant | 3.5101*** | 16.6485*** | 6.7777*** | 8.8657 | 0.6777*** | 1.8371*** | 9.5104*** | -4.4056 |
| | (0.0264) | (5.8209) | (0.0453) | (6.6099) | (0.0043) | (0.6469) | (0.0695) | (4.9362) |
| Observations | 106 | 106 | 104 | 104 | 104 | 104 | 106 | 106 |
| R-squared | 0.8718 | 0.9059 | 0.9852 | 0.9877 | 0.9753 | 0.9856 | 0.9765 | 0.9956 |
| Bank_Subsidiary FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter FE | NO | NO | NO | NO | NO | NO | NO | NO |

**Table 5**

**Cross-guarantee and subsidiary bank's risk-taking**. This table reports results using Equation (3). Models (1), (3), (5), and (7) contain Multi-bank, Parent Debt Ratio, and the interaction item as controls, while Models (2), (4), (6), and (8) contain all set of controls. All models include subsidiary bank and quarter fixed effects. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

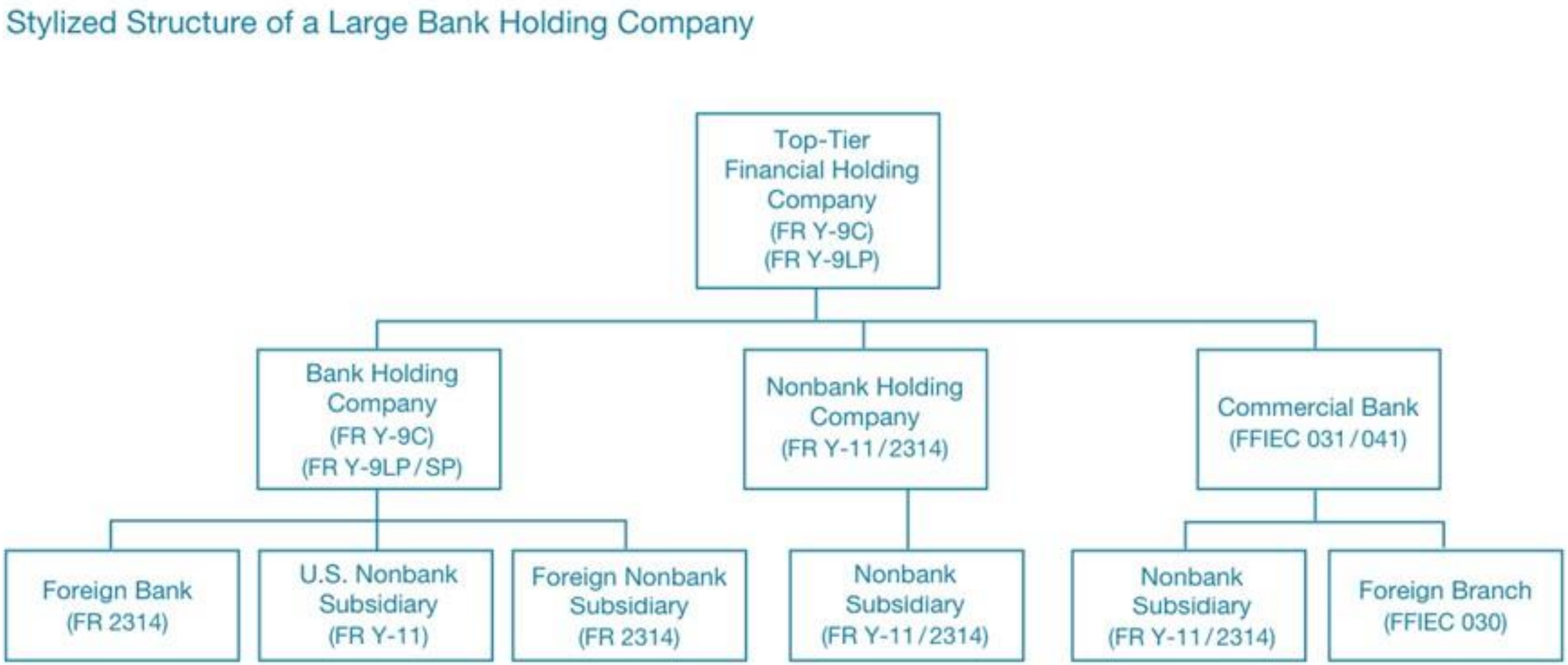
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| VARIABLES | Z-Score | Z-Score | RW Assets/T1 Capital | RW Assets/T1 Capital | RW Assets/Assets | RW Assets/Assets | Assets /T1 Capital | Assets /T1 Capital |
| Multi-bank X Parent Debt Ratio | 0.9618 | 0.6064 | -1.8327* | -1.6793* | -0.2331** | -0.1958** | 0.7193 | 0.7743 |
| | (0.6968) | (0.4941) | (0.9419) | (0.8909) | (0.1054) | (0.0980) | (1.5396) | (1.0675) |
| Parent Debt Ratio | 0.3409*** | 0.2460** | -1.2347*** | -0.8715** | 0.0779** | 0.0401 | -2.8962*** | -1.7352*** |
| | (0.1108) | (0.1234) | (0.4277) | (0.4282) | (0.0380) | (0.0362) | (0.6027) | (0.4852) |
| Multi-bank | (Omited) | | | | | | | |
| Tobin Q | | -0.3149 | | 1.5706 | | 0.2084** | | 0.8533 |
| | | (0.4102) | | (1.1126) | | (0.1005) | | (1.2686) |
| Ln AT | | -0.2507** | | 0.1664 | | 0.0014 | | 0.4405*** |
| | | (0.1143) | | (0.1372) | | (0.0175) | | (0.1562) |
| Debt/Equity | | -0.0601*** | | 0.0977*** | | -0.0100*** | | 0.3432*** |
| | | (0.0184) | | (0.0355) | | (0.0037) | | (0.0599) |
| Constant | 3.4452*** | 8.1097*** | 7.1220*** | 2.1745 | 0.6755*** | 0.5105 | 10.0740*** | -0.1932 |
| | (0.0394) | (2.0593) | (0.0570) | (2.6820) | (0.0060) | (0.3243) | (0.0897) | (3.0526) |
| Observations | 6,258 | 6,258 | 6,232 | 6,232 | 6,232 | 6,232 | 6,258 | 6,258 |
| R-squared | 0.6902 | 0.7216 | 0.9486 | 0.9507 | 0.9135 | 0.9173 | 0.9223 | 0.9351 |
| Bank_Subsidiary FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |

**Table 6**

**Parent debt ratio and subsidiary bank's risk-taking.** This table presents the results from the models for our preliminary analysis used in Section (7). The dependent variables are the logarithm of Z-score (Z-Score), Risk-weighted Leverage (RW Assets/T1 Capital), Asset Risk Weight (RW Assets/Assets) and Leverage (Assets /T1 Capital). The key explanatory variable is parent debt ratio. For each risk-taking measure, Models (1), (3), (5), and (7) contain only parent debt ratio as a control, while Models (2), (4), (6), and (8) contain all set of controls. All models use quarterly data for 2017-2022. All models include both subsidiary bank and quarter fixed effects. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

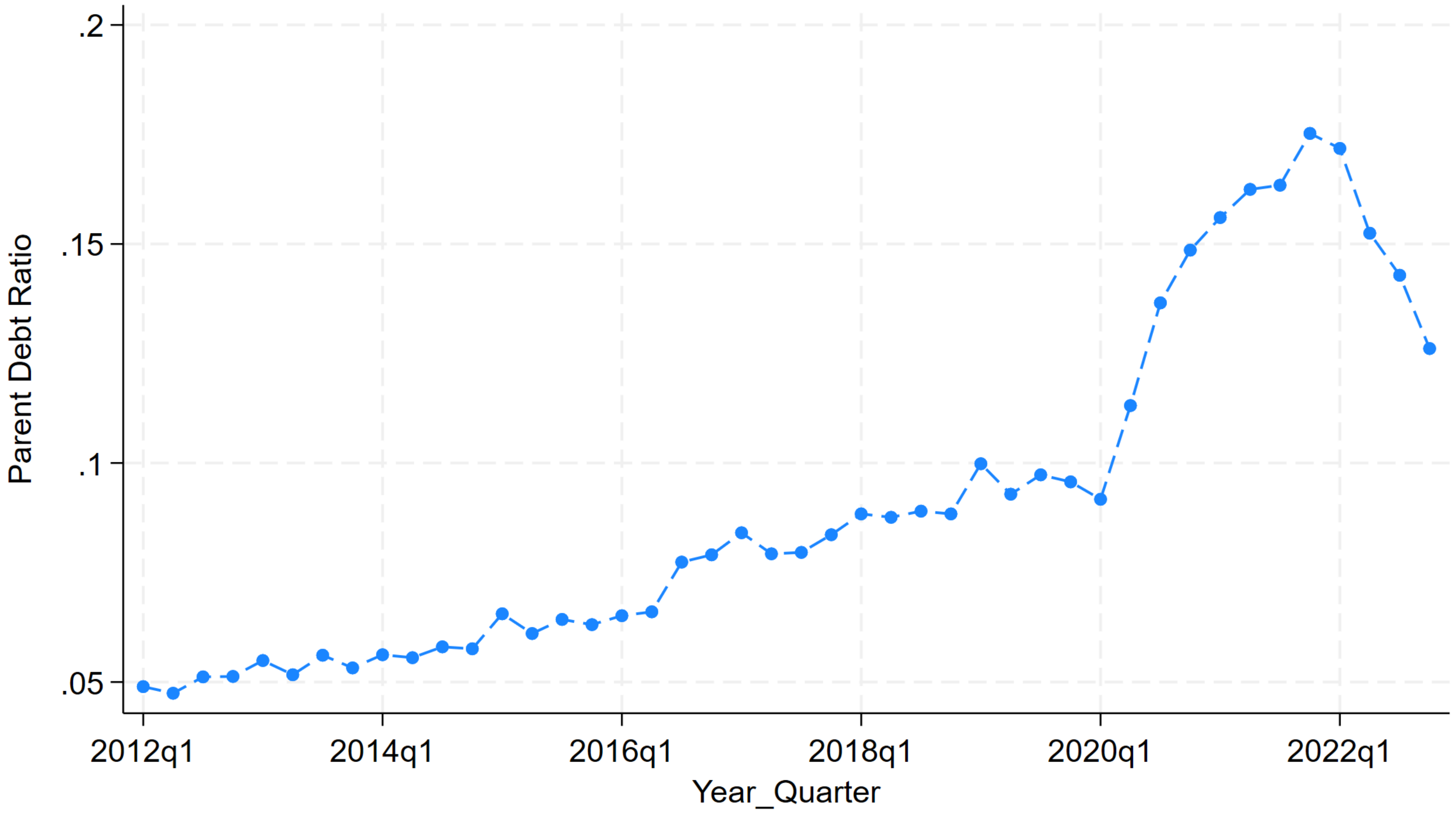
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| VARIABLES | Z-Score | Z-Score | RW Assets/T1 Capital | RW Assets/T1 Capital | RW Assets/Assets | RW Assets/Assets | Assets /T1 Capital | Assets /T1 Capital |
| Parent Debt Ratio | 0.4251*** | 0.3655** | -0.5952** | -0.5062* | 0.0801** | 0.0610* | -2.2042*** | -1.7158*** |
| | (0.1372) | (0.1470) | (0.2980) | (0.2784) | (0.0366) | (0.0318) | (0.5298) | (0.5068) |
| Tobin Q | | 0.1415 | | 1.6096 | | 0.2610*** | | -0.2883 |
| | | (0.4189) | | (1.1477) | | (0.0945) | | (1.6805) |
| Ln AT | | -0.3066* | | -0.2339** | | -0.0208 | | -0.0920 |
| | | (0.1819) | | (0.1159) | | (0.0128) | | (0.1927) |
| Debt/Equity | | -0.0257* | | 0.0294 | | -0.0081** | | 0.1931* |
| | | (0.0141) | | (0.0292) | | (0.0033) | | (0.0995) |
| Constant | 3.4015*** | 8.2585*** | 6.8700*** | 8.6536*** | 0.6531*** | 0.7684*** | 10.0370*** | 10.3684*** |
| | (0.0180) | (2.9907) | (0.0394) | (2.2297) | (0.0048) | (0.2378) | (0.0695) | (3.4460) |
| Observations | 3,136 | 3,136 | 3,110 | 3,110 | 3,110 | 3,110 | 3,136 | 3,136 |
| R-squared | 0.7865 | 0.7967 | 0.9745 | 0.9749 | 0.9524 | 0.9547 | 0.9474 | 0.9490 |
| Bank_Subsidiary FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |

**Figure 1: The organizational structure of a large BHC**



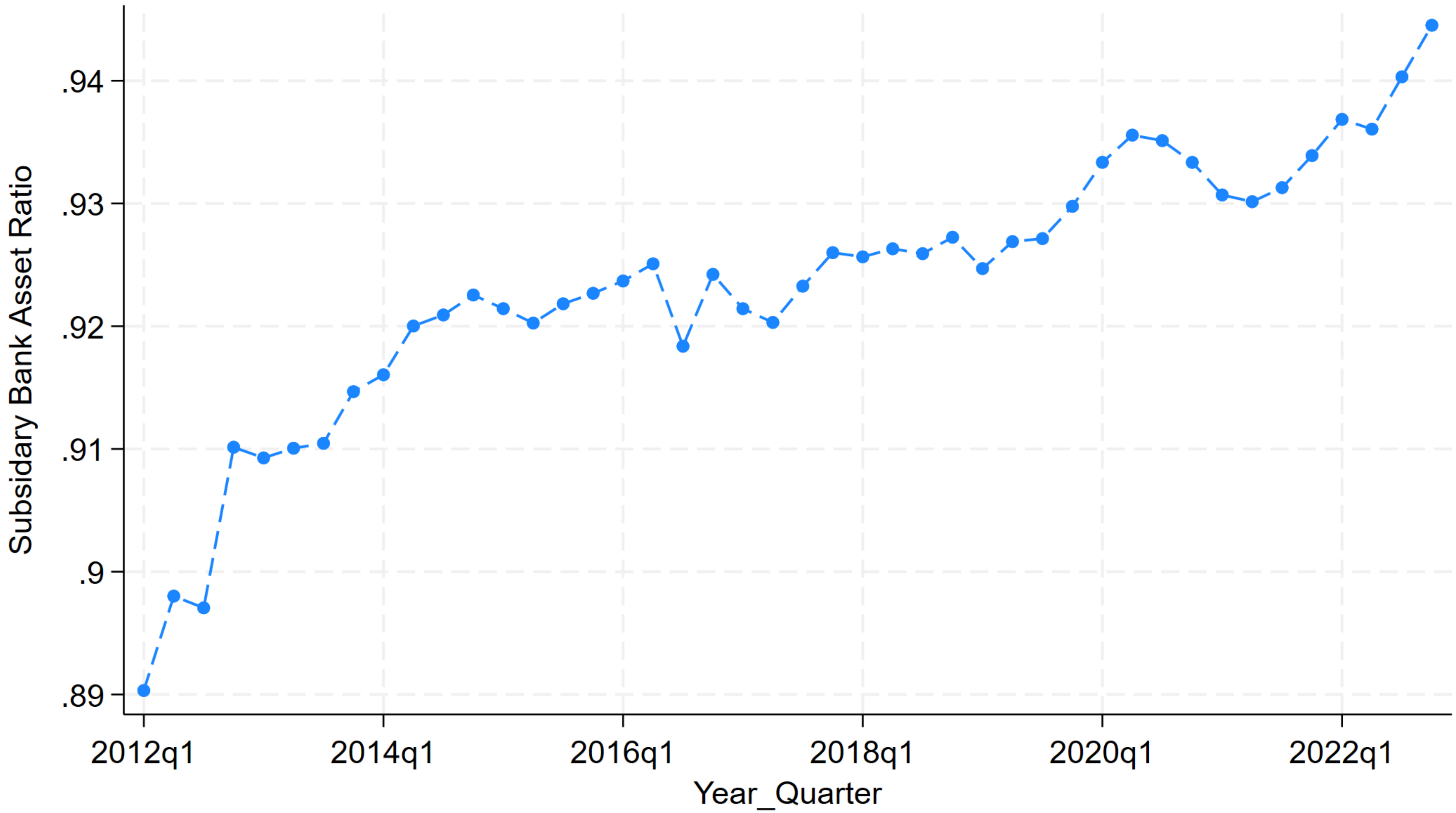

Data sources: Avraham et al. (2012)

**Figure 2: Parent debt ratio over time**




Source: FR Y-9C

**Figure 3: Subsidiary bank's asset ratio over time**




Source: FR Y-9C and Call Reports

# Appendix A: Proofs of Lemmas

# Appendix A1: Proof of Lemma 1

Lemma 1: If and only if $\alpha > \frac{(1-E_p)-S}{R_n-S} \equiv \widehat{\alpha}$, it holds that $\widehat{D}'_p < \widehat{D}''_p$. Excessive risk-taking occurs when $D_p < \widehat{D}'_p$, whereas the first-best risk level emerges when $D_p \in \left[\widehat{D}'_p, \widehat{D}''_p\right]$.

Proof. Based on Equations (7) and (8), when $S(1-\alpha) - D_b < 0$, the subsidiary bank fails, implying that $D_p < \widehat{D}'_p$ . The BHC shareholder's payoff is:

$$L = P(R_b)[R_n\alpha - D_n + R_b(1-\alpha) - D_b - D_p] + (1 - P(R_b)[R_n\alpha - D_n - D_p] \quad (1)$$

where $D_b = (1-\alpha)(1-E_p-D_p)$, and $D_n = \alpha(1-E_p-D_p)$. Taking the first-order derivative:

$-\beta R_b(1-\alpha) + (1-\beta R_b)(1-\alpha) + \beta(1-\alpha)(1-E_p-D_p) = 0$

$1 + \beta(1-E_p-D_p) = 2\beta R_b$

$R_b = \frac{1+\beta(1-E_p-D_p)}{2\beta}$

When $S(1-\alpha)-D_b > 0$, the subsidiary bank succeeds, implying that $D_p \in \left[\widehat{D}'_p, \widehat{D}''_p\right]$. The BHC shareholder's payoff is:

$$L = P(R_b)[R_n\alpha - D_n + R_b(1-\alpha) - D_b - D_p] + (1 - P(R_b)[R_n\alpha - D_n + S(1-\alpha) - D_b - D_p] \quad (2)$$

Taking the first-order condition is

$-\beta R_b(1-\alpha) + (1-\beta R_b)(1-\alpha) + \beta S(1-\alpha) = 0$

$-\beta R_b + (1-\beta R_b) + \beta S = 0$

$-\beta R_b + (1-\beta R_b) + \beta S = 0$

$R_b = \frac{1+\beta S}{2\beta}$

## Appendix A2: Proof of Lemma 4

Lemma 4: When both the parent/BHC and the subsidiary bank fail, excessive risk-taking still occurs because limited liability distortion emerges at the subsidiary bank level.

Proof. If parent/BHC fails, market discipline exists on the parent debt, and the interest rate of parent debt is $r_p$ instead of risk-free rate. When the parent/BHC fails, the equity value of non-bank subsidiary is insufficient to cover the parent debt obligations with

$$R_n\alpha - \alpha(1 - E_p - D_p) - r_p * D_p < 0 \tag{3}$$

When the subsidiary bank fails, the asset of the subsidiary bank does not cover its debt

$$S(1 - \alpha) < (1 - \alpha)(1 - E_p - D_p) \tag{4}$$

The creditor's break-even condition is

$$r_p D_p(1 - \beta R_b) + \beta R_b[R_n\alpha - \alpha(1 - E_p - D_p)] = D_p \tag{5}$$

The BHC shareholder's payoff is

$$(1 - \beta R_b)[R_n\alpha + R_b(1 - \alpha) - (1 - E_p - D_p) - r_p D_p] + \beta R_b \cdot 0 \tag{6}$$

After rearranging, this simplifies to

$(1 - \beta R_b)[R_n\alpha + R_b(1 - \alpha) - (1 - E_p - D_p)] - D_P + \beta R_b[R_n\alpha - \alpha(1 - E_p - D_p)]$

Taking the first-order condition:

$-\beta R_b(1 - \alpha) + (1 - \beta R_b)(1 - \alpha) + \beta(1 - \alpha)(1 - E_p - D_p) = 0$

$1 + \beta(1 - E_p - D_p) = 2\beta R_b$

Solving for $R_b$

$R_b = \frac{1+\beta(1-E_p-D_p)}{2\beta}$

## Appendix A3: Proof of Lemma 5

Lemma 5: When the parent/BHC fails but the subsidiary bank does not, it yields the first best solution with

$$R_b^{FB} = \frac{1+\beta S}{2\beta}. \tag{7}$$

Proof. The parent/BHC fails only if the subsidiary bank fails in its investment, which occurs with probability $\beta R_b$. Since market discipline exists in the parent debt, the interest rate on parent debt is $r_p$.

When the parent/BHC fails, it requires

$$[R_n\alpha - \alpha(1-E_p-D_p)] + [S(1-\alpha) - (1-\alpha)(1-E_p-D_p)] - r_pD_p < 0 \tag{8}$$

When the subsidiary bank does not fail

$$[R_n\alpha - \alpha(1-E_p-D_p)] + [R_b(1-\alpha) - (1-\alpha)(1-E_p-D_p)] - r_pD_p > 0 \tag{9}$$

The BHC shareholder's payoff is

$$(1-\beta R_b)[R_n\alpha + R_b(1-\alpha) - (1-E_p-D_p) - r_pD_p] + \beta R_b * 0 \tag{10}$$

While the loss from risk-taking is fully passed onto the parent, limited liability at the parent level exists. But market discipline should counter this effect.

The creditor's break-even condition is

$$r_pD_p(1-\beta R_b) + \beta R_b[R_n\alpha + S(1-\alpha) - (1-E_p-D_p)] = D_p \tag{11}$$

The BHC shareholder's payoff is

$$(1-\beta R_b)[R_n\alpha + R_b(1-\alpha) - (1-E_p-D_p) - r_pD_p] + \beta R_b \cdot 0 \tag{12}$$

After rearranging, it is equal to

$$(1-\beta R_b)[R_n\alpha+R_b(1-\alpha)-(1-E_p-D_p)-\frac{D_p-\beta R_b[R_n\alpha+S(1-\alpha)-(1-E_p-D_p)]}{(1-\beta R_b)}] \tag{13}$$

$$= (1-\beta R_b)[R_n\alpha+R_b(1-\alpha)-(1-E_p-D_p)]-D_p+\beta R_b[R_n\alpha+S(1-\alpha)-(1-E_p-D_p)]$$

$$= R_n\alpha + (1-\beta R_b)R_b(1-\alpha) + \beta R_b S(1-\alpha) - (1-E_p-D_p) - D_p$$

Taking the first-order condition is

$$-\beta R_b(1-\alpha) + (1-\beta R_b)(1-\alpha) + \beta S(1-\alpha) = 0$$

$$-\beta R_b + (1-\beta R_b) + \beta S = 0$$

$$-\beta R_b + (1-\beta R_b) + \beta S = 0$$

$R_b = \frac{1+\beta S}{2\beta}$